

Day 32: Azure Data Factory (ADF)

Curriculum Notes & Step-by-Step Implementation Guide

1. Introduction to Azure Data Factory (ADF)

Azure Data Factory is Microsoft Azure's cloud-based ETL (Extract, Transform, Load) and data integration service. It allows you to create data-driven workflows for orchestrating data movement and transforming data at scale. Rather than storing data, it acts as an "orchestrator" or conductor that coordinates external systems (like SQL Databases, Data Lakes, or Spark clusters) to execute the work.

2. Core Topics & Architecture Concepts

Pipelines & Activities

A **Pipeline** is a logical grouping of activities that performs a unit of work. Together, the activities in a pipeline perform a complete task.

An **Activity** represents a processing step within a pipeline. ADF supports three main types of activities:

- **Data movement:** E.g., the Copy Activity, which copies data from a source to a sink.
- **Data transformation:** E.g., Data Flows, executing a Databricks Notebook, or running a Stored Procedure.
- **Control flow:** E.g., ForEach loops, If Conditions, or executing other pipelines.

Datasets & Linked Services

Linked Services act like connection strings. They define the connection information needed for ADF to connect to external data sources or compute environments (e.g., credentials to access an Azure Blob Storage account).

Datasets represent the structure of the data within those linked data stores (e.g., specifying that the data is a CSV file located in a specific folder).

Mapping Data Flows

Data Flows allow data engineers to develop graphic data transformation logic without writing code. The resulting data flows are executed as activities within ADF pipelines on scaled-out Apache Spark clusters, making them highly efficient for big data transformation.

Triggers

Triggers represent the unit of processing that determines when a pipeline execution needs to be kicked off. They can be scheduled (time-based), event-driven (e.g., reacting to a file arriving in blob storage), or based on tumbling windows.

3. Step-by-Step Implementation Guide in Azure

Phase 1: Setting up the Azure Data Factory

Step 1: Create the ADF Resource

1. Log in to the **Azure Portal** (portal.azure.com).
2. In the search bar, type "Data factories" and select the service.
3. Click **+ Create**.
4. Select your **Subscription** and **Resource Group**.
5. Provide a globally unique **Name** for the factory and select your target **Region**.
6. Under the "Git configuration" tab, check **Configure Git later** (for a simpler initial setup).
7. Click **Review + Create**, wait for validation, and then click **Create**.
8. Once deployment is complete, navigate to the resource and click **Launch studio** to open the Azure Data Factory UI.

Phase 2: Building a Pipeline with Copy Activities

Step 2: Create Linked Services (Connections)

1. In ADF Studio, navigate to the **Manage** tab (toolbox icon on the far left).
2. Under the "Connections" section, select **Linked services** and click **+ New**.
3. Select your source data store (e.g., Azure Data Lake Storage Gen2) and click Continue.
4. Name the service, select the Authentication method, and choose the storage account. Click **Create**.
5. Repeat this process to create a Linked Service for your destination (e.g., Azure SQL Database).

Step 3: Define Datasets

1. Switch to the **Author** tab (pencil icon).
2. Hover over "Datasets", click the ... (or the + icon) and select **New dataset**.
3. Select your source data store format (e.g., DelimitedText for CSV in ADLS Gen2).
4. Select the Source Linked Service you created in Step 2. Browse to select the specific file path and check "First row as header" if applicable. Click OK.
5. Repeat this process to create a Sink (Destination) Dataset linked to your target SQL table.

Step 4: Create the Pipeline and Copy Activity

1. Under the **Author** tab, click the + icon and select **Pipeline**.
2. In the Activities pane, expand "Move and transform" and drag the **Copy data** activity onto the canvas.
3. Select the Copy activity on the canvas to configure it in the bottom properties pane:
 - **Source tab:** Select your Source Dataset.
 - **Sink tab:** Select your Sink Dataset.
 - **Mapping tab:** Click "Import schemas" to map the source columns to the destination database columns.
4. Click **Debug** at the top of the canvas to test the pipeline run manually.

Phase 3: Building Transformations using Data Flows

Step 5: Create a Mapping Data Flow

1. In the **Author** tab, click + -> **Data Flow**.
2. Toggle on **Data flow debug** at the top to spin up a Spark cluster for interactive testing.
3. Click **Add Source** on the canvas and point it to your source dataset.
4. Click the + icon next to the source step to add transformation modules:
 - *Filter*: To remove rows based on conditions.
 - *Derived Column*: To create new columns or transform existing ones.
 - *Aggregate*: To group by and calculate metrics.
5. Finally, add a **Sink** transformation and map it to your target dataset.
6. Return to your main pipeline, drag a **Data flow** activity onto the canvas, and link it to this new Data Flow.

Phase 4: Automation

Step 6: Configure Triggers & Publish

1. To schedule the pipeline, open the pipeline canvas and click **Add trigger** -> **New/Edit**.
2. Click + **New** in the dropdown menu.
3. Name your trigger and select the type (e.g., **Schedule** for daily runs, or **Storage Events** to run when a new file arrives).
4. Configure the execution schedule and click OK.

Crucial Final Step: Click **Publish all** at the top of ADF Studio. ADF saves your work in the browser until you publish. Triggers will not activate and pipelines won't be saved to the live service until you successfully publish.